Ordinary Differential Equations

A First-Principles Cheat Sheet for Four Core Methods

Color key: all live solution steps in the **Worked Examples** are typeset in **purple**; explanatory text stays black. A general first-order ODE is written

$$\frac{dy}{dx} = f(x, y),$$

and each method below is simply a different "shape" that $f(x, y)$ can take.

1. Separable Equations

The Core Logic

An equation is **separable** when the right-hand side factors into a function of x alone times a function of y alone:

$$\frac{dy}{dx} = g(x)\,h(y).$$

Splitting the differentials looks like fraction algebra, but it is really the *substitution rule (reverse chain rule)*. Writing $\frac{1}{h(y)}\frac{dy}{dx} = g(x)$ and integrating both sides with respect to x,

$$\int \frac{1}{h(y)}\frac{dy}{dx}\,dx = \int g(x)\,dx,$$

the left side becomes $\int \frac{1}{h(y)}\,dy$ because $\frac{dy}{dx}\,dx = dy$. That is *why* separating the differentials is legitimate.

All Possible Cases / Variations

- **Explicit vs. implicit.** Sometimes you can isolate y; other times the cleanest answer is a relation $F(x, y) = C$.
- **Equilibrium / lost solutions.** Dividing by $h(y)$ is illegal where $h(y) = 0$. Each root $y = c$ is a constant solution that may be lost — check it separately.
- **Disguised separability.** Factoring may be hidden: e.g. $\frac{dy}{dx} = x + xy = x(1 + y)$ or $\frac{dy}{dx} = e^{x+y} = e^x e^y$.
- **Non-elementary integrals.** One side may have no closed-form antiderivative; leave it as an integral.
- **Initial Value Problem (IVP).** A condition $y(x_0) = y_0$ fixes the constant C.

Step-by-Step Methodology

1. Write the equation as $\frac{dy}{dx} = g(x)\,h(y)$.
2. Solve $h(y) = 0$ for the equilibrium solutions; set them aside.
3. Separate: $\frac{dy}{h(y)} = g(x)\,dx$.
4. Integrate both sides.
5. Combine the two constants into a single C.

6. Solve for y if possible (explicit); otherwise leave implicit.
7. Apply the initial condition to find C.
8. State equilibrium solutions and the domain of validity.

Worked Example A — Explicit IVP

Solve $\dfrac{dy}{dx} = 2xy, \quad y(0) = 1.$

$$\begin{aligned}
\frac{dy}{dx} &= 2xy && \text{(note: } y = 0 \text{ is an equilibrium solution)} \\
\tfrac{1}{y}\,dy &= 2x\,dx && \text{(separate variables)} \\
\int \tfrac{1}{y}\,dy &= \int 2x\,dx && \text{(integrate both sides)} \\
\ln|y| &= x^2 + C_1 && \\
|y| &= e^{x^2 + C_1} = e^{C_1} e^{x^2} && \\
y &= Ce^{x^2} && \text{(let } C = \pm e^{C_1}\text{)}
\end{aligned}$$

Apply $y(0) = 1$:

$$1 = Ce^0 = C \qquad \Longrightarrow \qquad \boxed{y = e^{x^2}}$$

Worked Example B — Implicit Solution

Solve $\dfrac{dy}{dx} = \dfrac{3x^2 + 4x + 2}{2(y-1)}, \quad y(0) = -1.$

$$\begin{aligned}
2(y-1)\,dy &= (3x^2 + 4x + 2)\,dx && \text{(separate)} \\
\int 2(y-1)\,dy &= \int (3x^2 + 4x + 2)\,dx && \text{(integrate)} \\
y^2 - 2y &= x^3 + 2x^2 + 2x + C &&
\end{aligned}$$

Apply $y(0) = -1$:

$$(-1)^2 - 2(-1) = C \qquad \Longrightarrow \quad 1 + 2 = C \;\Rightarrow\; C = 3$$

$$\boxed{y^2 - 2y = x^3 + 2x^2 + 2x + 3} \qquad \text{(implicit solution)}$$

Solving the quadratic in y for an explicit branch:

$$\begin{aligned}
y &= 1 \pm \sqrt{x^3 + 2x^2 + 2x + 4} \\
y &= 1 - \sqrt{x^3 + 2x^2 + 2x + 4} && (-\text{ branch matches } y(0) = -1)
\end{aligned}$$

2. Linear First-Order Equations (Integrating Factors)

The Core Logic

A linear first-order ODE in **standard form** is

$$\frac{dy}{dx} + P(x)\,y = Q(x).$$

The left side is almost a product-rule derivative. Multiply by an **integrating factor** $\mu(x)$ so that

$$\mu\frac{dy}{dx} + \mu P\,y \;=\; \frac{d}{dx}(\mu y) = \mu\frac{dy}{dx} + \frac{d\mu}{dx}\,y.$$

Matching the y-terms forces $\dfrac{d\mu}{dx} = \mu P$, a separable equation giving

$$\mu(x) = e^{\int P(x)\,dx}.$$

The factor *manufactures* a perfect product-rule derivative, so one integration finishes the job.

All Possible Cases / Variations

- **Not in standard form.** Divide by the coefficient of y' first; beware points where it vanishes (singular points).
- **Homogeneous $Q(x) = 0$.** The equation is then also separable.
- **Constant coefficient.** If P is constant, $\mu = e^{Px}$.
- **Drop the constant & bars.** When integrating P, omit $+C$ and absolute values — any valid μ works.
- **Linear in x, not y.** Rewrite as $\dfrac{dx}{dy} + P(y)\,x = Q(y)$ and use $\mu(y)$.
- **Bernoulli.** $y' + Py = Qy^n$ becomes linear under the substitution $v = y^{1-n}$.

Step-by-Step Methodology

1. Put in standard form $y' + P(x)y = Q(x)$.
2. Compute $\mu(x) = e^{\int P\,dx}$ (drop the constant).
3. Multiply every term by $\mu(x)$.
4. Recognize the left side as $\frac{d}{dx}\left[\mu y\right]$.
5. Integrate: $\mu y = \int \mu Q\,dx + C$.
6. Solve for y.
7. Apply the initial condition for C.

Worked Example — IVP

Solve $\;x\dfrac{dy}{dx} + 2y = 4x^2, \quad y(1) = 2.$

Step 1 — Standard form (divide by x):

$$\frac{dy}{dx} + \tfrac{2}{x}\,y = 4x, \qquad P(x) = \tfrac{2}{x}, \quad Q(x) = 4x.$$

Step 2 — Integrating factor:

$$\mu(x) = e^{\int \frac{2}{x}\,dx} = e^{2\ln|x|} = e^{\ln x^2} = x^2.$$

Steps 3–4 — Multiply by $\mu = x^2$ and collapse the left side:

$$\begin{aligned} x^2\frac{dy}{dx} + 2x\,y &= 4x^3 \\ \frac{d}{dx}\left(x^2 y\right) &= 4x^3 && \text{(exact product-rule derivative)} \end{aligned}$$

Step 5 — Integrate both sides:

$$x^2 y = \int 4x^3\,dx = x^4 + C$$

Step 6 — Solve for y:

$$y = x^2 + \frac{C}{x^2}$$

Step 7 — Apply $y(1) = 2$:

$$2 = 1 + C \implies C = 1 \qquad \implies \qquad \boxed{y = x^2 + \tfrac{1}{x^2}}$$

3. Exact Equations

The Core Logic

In **differential form** $M(x,y)\,dx + N(x,y)\,dy = 0$ the equation is **exact** when the left side is the total differential of a hidden potential $F(x,y)$:

$$dF = F_x\,dx + F_y\,dy = 0, \qquad F_x = M, \quad F_y = N.$$

Then $F(x,y) = C$ along every solution curve. By Clairaut's theorem ($F_{xy} = F_{yx}$),

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \iff \quad \text{the equation is exact.}$$

All Possible Cases / Variations

- **Exactness test.** If $M_y = N_x$, integrate directly; if $M_y \neq N_x$, seek an integrating factor.
- **Integrating factor for non-exact equations:**
  - If $\dfrac{M_y - N_x}{N}$ depends on x only, then $\mu(x) = e^{\int \frac{M_y - N_x}{N}\,dx}$.
  - If $\dfrac{N_x - M_y}{M}$ depends on y only, then $\mu(y) = e^{\int \frac{N_x - M_y}{M}\,dy}$.
- **Implicit answers are normal.** $F(x,y) = C$ often cannot be solved explicitly for y.
- **Rearrangement.** You may need to reorder terms into $M\,dx + N\,dy = 0$.
- **IVP.** Use $y(x_0) = y_0$ to evaluate C.

Step-by-Step Methodology

1. Arrange as $M\,dx + N\,dy = 0$; identify M, N.
2. Test exactness: $M_y \overset{?}{=} N_x$. (If not, find μ and restart.)
3. Integrate M in x: $F = \int M\,dx + g(y)$.

4. Set $F_y = N$ and solve for $g'(y)$.
5. Integrate $g'(y)$ to get $g(y)$.
6. Write the solution $F(x, y) = C$.
7. Apply the initial condition for C.

Worked Example A — Directly Exact

Solve $(2xy + 3)\,dx + (x^2 - 1)\,dy = 0$.

Identify & test:

$$\begin{aligned} M &= 2xy + 3, & N &= x^2 - 1 \\ M_y &= 2x, & N_x &= 2x \implies \text{exact.} \end{aligned}$$

Integrate M in x:

$$F(x, y) = \int (2xy + 3)\,dx + g(y) = x^2 y + 3x + g(y).$$

Match F_y to N:

$$F_y = x^2 + g'(y) \stackrel{!}{=} x^2 - 1 \implies g'(y) = -1 \implies g(y) = -y.$$

Solution:

$$\boxed{x^2 y + 3x - y = C}$$

Worked Example B — Non-Exact, Needs an Integrating Factor

Solve $(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$.

Test exactness:

$$\begin{aligned} M &= 3xy + y^2, & N &= x^2 + xy \\ M_y &= 3x + 2y, & N_x &= 2x + y \implies \text{NOT exact.} \end{aligned}$$

Try the x-only criterion:

$$\frac{M_y - N_x}{N} = \frac{(3x + 2y) - (2x + y)}{x^2 + xy} = \frac{x + y}{x(x + y)} = \frac{1}{x}$$

which depends on x only, so

$$\mu(x) = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x.$$

Multiply through by $\mu = x$:

$$(3x^2 y + xy^2)\,dx + (x^3 + x^2 y)\,dy = 0$$

Re-test on M^*, N^*:

$$M^*_y = 3x^2 + 2xy, \qquad N^*_x = 3x^2 + 2xy \checkmark \text{ (now exact)}$$

Integrate and match:

$$F = \int (3x^2y + xy^2)\,dx + g(y) = x^3y + \tfrac{1}{2}x^2y^2 + g(y)$$

$$F_y = x^3 + x^2y + g'(y) \overset{!}{=} x^3 + x^2y \implies g'(y) = 0$$

$$\boxed{x^3y + \tfrac{1}{2}x^2y^2 = C}$$

4. Numerical Approximation — Euler's Method

The Core Logic

For an IVP $y' = f(x,y),\ y(x_0) = y_0$ with no closed form, Euler's method walks along tangent lines. The first-order Taylor expansion gives, for a small step h,

$$y(x+h) \approx y(x) + h\,y'(x) = y(x) + h\,f\big(x, y(x)\big).$$

We read the slope $f(x,y)$ from the equation itself, take a short straight step, and repeat — trading a little accuracy (the curve bends off the tangent) for the ability to march forward without integrating.

All Possible Cases / Variations

- **Step size h given** vs. **n steps over** $[a,b]$, where $h = \frac{b-a}{n}$.
- **Forward (explicit) Euler** is the standard form used here.
- **Error behavior.** Local error per step is $O(h^2)$; global error is $O(h)$ — halving h roughly halves total error.
- **Comparison to exact.** When available, tabulate the exact value to report the error.
- **Better cousins.** Improved Euler / Heun (predictor–corrector) and Runge–Kutta (RK4) cut error sharply for the same h.

Step-by-Step Methodology

1. Identify $f(x,y)$, the start (x_0, y_0), the step h, and the target x.
2. Iterate: $x_{n+1} = x_n + h, \quad y_{n+1} = y_n + h\,f(x_n, y_n)$.
3. Evaluate the slope $f(x_n, y_n)$ at the current point.
4. Step forward; record (x_{n+1}, y_{n+1}) in a table.
5. Repeat to the target x.
6. (Optional) Compare to the exact solution for the error.

Worked Example

Approximate $y(0.3)$ for $\dfrac{dy}{dx} = x + y,\ y(0) = 1,$ using $h = 0.1$.

Here $f(x,y) = x + y$, $x_0 = 0$, $y_0 = 1$, and $y_{n+1} = y_n + h(x_n + y_n)$.

Step 1 $(n = 0)$:

$$f(x_0, y_0) = 0 + 1 = 1$$

$$y_1 = 1 + (0.1)(1) = 1.100, \qquad x_1 = 0.1$$

Step 2 ($n = 1$):

$$f(x_1, y_1) = 0.1 + 1.100 = 1.200$$
$$y_2 = 1.100 + (0.1)(1.200) = 1.220, \qquad x_2 = 0.2$$

Step 3 ($n = 2$):

$$f(x_2, y_2) = 0.2 + 1.220 = 1.420$$
$$y_3 = 1.220 + (0.1)(1.420) = 1.362, \qquad x_3 = 0.3$$

Result: $\boxed{y(0.3) \approx 1.362}$

n	x_n	y_n	$f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + hf$
0	0.0	1.000	1.000	1.100
1	0.1	1.100	1.200	1.220
2	0.2	1.220	1.420	1.362
3	0.3	1.362	—	—

Accuracy check. The exact solution is $y = 2e^x - x - 1$, so

$$y_{\text{exact}}(0.3) = 2e^{0.3} - 1.3 \approx 1.39972$$
$$\text{Error} = |\, 1.39972 - 1.362 \,| \approx 0.038.$$

The estimate undershoots because the true curve is concave up — the tangent lines fall below it. A smaller h (or Heun/RK4) shrinks the gap.

5. Quick-Reference Decision Map

Given $\dfrac{dy}{dx} = f(x, y)$ — or $M\,dx + N\,dy = 0$ — ask in order:

1. **Does f factor as $g(x)\,h(y)$?** → **Separable** (separate & integrate).
2. **Can it be written $y' + P(x)y = Q(x)$?** → **Linear** (factor $\mu = e^{\int P\,dx}$).
3. **Form $M\,dx + N\,dy = 0$ with $M_y = N_x$?** → **Exact** (find F, set $F = C$). If $M_y \neq N_x$, use an integrating factor $\mu(x)$ or $\mu(y)$ to make it exact.
4. **No closed form / need a number?** → **Euler's Method** (step along tangents).

Many equations fit more than one category (a homogeneous linear equation is also separable) — pick the cleanest path.